

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Hackathon

Course Code:	DSA0404	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO5	Assessment:	Assessment Tool 3- Hackathon
Weightage:	40% of CIA	Total Marks:	100
CO3 Statement:	Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)		
Student Name:	B C Yugesh	Reg. No.:	192424282
Section / Batch:	2024	Date:	31/08/2026

Code of Conduct

I B C Yugesh (192424282) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: B C YUGESH

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	

Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Problem Statement:

House Price Prediction. Predict housing prices from features like area, location, number of rooms, and age of property. Students evaluate using RMSE, MAE, and R^2 .

localhost:8501

Deploy



House Price Prediction System

Predict house prices using Area, Location, Number of Rooms, and Property Age.



Model Performance

RMSE	MAE	R^2 Score
₹639,417	₹476,368	0.9595

localhost:8501

Deploy



Enter House Details

Area (sq.ft)

1500

- +

Location

Bangalore

▼

Number of Rooms

3

- +

Property Age (years)

5

- +

 Predict House Price

House Price Prediction

localhost:8501

Deploy

Dataset Preview

	Area	Location	Rooms	Age	Price	
0	800	Chennai		2	20	4200000
1	900	Chennai		2	15	4800000
2	1000	Chennai		2	12	5200000
3	1100	Chennai		2	10	5800000
4	1200	Chennai		3	8	6500000
5	1300	Chennai		3	6	7000000
6	1400	Chennai		3	5	7600000
7	1500	Chennai		3	4	8200000
8	1600	Chennai		3	3	8800000
9	1800	Chennai		4	2	9800000

File Edit Selection View Go Run Terminal Help

House_Price_Prediction

app.py model.py house_data.csv

```
44 X_train, X_test, y_train, y_test = train_test_split(
45     X,
46     y,
47     test_size=0.2,
48     random_state=42
49 )
50
51 # -----
52 # Train Model
53 # -----
54 model = LinearRegression()
55 model.fit(X_train, y_train)
56
57 # -----
58 # Model Evaluation
59 # -----
60 y_pred = model.predict(X_test)
61
```

Actual Prices:

[6500000 6200000 8200000 4200000 6100000 15000000 11000000 10500000
 4500000 15000000 7600000 7900000 9800000 10000000 7000000]

Predicted Prices:

[6314378.81 6061565.68 8090707.52 3103768.33 5970782.38 14994143.24
11972735.61 11629139.18 3526001.67 14620294.84 7744451.17 8030863.7
10647578.51 10882030.15 7020364.76]

===== MODEL EVALUATION =====

RMSE: 639416.96

MAE: 476368.04

R² Score: 0.9595

Set BYOK utility models

Unlocks full AI features.

Configure

app.py

Describe what to build

Agent Models

Local Default permissions

Ln 186, Col 54 Spaces: 4 UTF-8 CRLF Python venv (3.13.15.final.0) Go Live

Geotag Photo:

